

Vani Rachanakar: A Simple AI-Assisted Utility for Creating Customised ESL Listening Materials

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Generative AI and neural speech synthesis have made it technically easy to produce high-quality speech from text. However, making this capability genuinely useful in a classroom still requires simple tools that fit naturally into a teacher's workflow. This is particularly relevant to English as a Second Language (ESL) education, where listening activities can expose learners to voices and accents that may be unfamiliar to them. Teachers may also wish to create conversations involving several speakers, with different voices and speaking rates, without having to learn audio-editing or other specialised software.

There are already text-to-speech solutions that allow different voices to be associated with different sections of a text. However, these systems can provide considerably more functionality than is required for the simple task of preparing a classroom listening activity. Vani Rachanakar was developed from the premise that a lightweight, teacher-oriented utility may be sufficient for this purpose.

Vani Rachanakar allows an ESL teacher to create customised listening conversations by selecting a voice and speaking rate for each individual dialogue entry. The teacher can input a conversation of their choice, assign a voice and speaking rate to each paragraph, listen to individual entries while making these choices, and finally generate a completely self-contained HTML page containing the conversation and its audio. Internet access is required only during the creation of the page, when the text is converted to speech. The resulting resource can subsequently be used by students without an Internet connection. The generated files are also relatively compact; a conversation containing 15 dialogue entries typically produces an HTML file of approximately 3 MB. Vani Rachanakar is currently available free of charge. The application can be accessed at <https://zeal-edu.co.in/VaniRachanakar/main.html>, with the source code openly available at <https://github.com/zeal-gaia/Vani-Rachanakar>.

The teacher has two options for providing the text of the conversation. A conversation can either be supplied as a text file, with individual dialogue entries separated by blank lines, or entered manually through the application. Once a paragraph has been entered, the teacher selects a voice and speaking rate and can test the resulting speech. Changing speaking rate may be useful for students who are new learners. The process of selecting and testing of voices and pace is inexpensive in terms of API calls and latency, as only one paragraph is converted. Also, the teacher can make out exactly how that specific paragraph sounds. Once all selections are over, the application generates a single HTML page which includes the text, the text converted to speech and the JavaScript code needed to display the text and play the speech. The page displays the dialogue entries in a visually differentiated format and plays the corresponding

speech sequentially. The page can be saved locally using standard browser controls and transferred to other devices as needed. No software other than a standard web browser is required to replay the conversation. The application currently uses Indian voices provided through Sarvam's text-to-speech service.

The resulting resource is intended to provide teachers with a simple way of adapting listening material to the needs of their learners. For example, a teacher may prepare a conversation involving three or four participants, assign a different voice to each speaker, and reduce the speaking rate for learners who require additional listening support. The same conversation can subsequently be reused without requiring the teacher or students to access the speech-synthesis service again.

The design of Vani Rachanakar deliberately emphasises simplicity. The objective was not to develop another sophisticated AI system, but to make an existing AI capability genuinely useful to a teacher. The teacher remains responsible for selecting the content, voices and speaking rates, while the application handles the otherwise technical task of generating and packaging the audio. This separation allows the technology to remain largely invisible to the teacher.

The utility is particularly relevant to inclusion and equitable access to learning resources in contexts where both linguistic familiarity and Internet availability can affect students' access to listening activities. Listening materials can be adapted to the linguistic context of learners rather than requiring all learners to work with a fixed recording. The availability of different Indian voices also provides an alternative to relying exclusively on voices and accents that may be unfamiliar to Indian ESL learners. The offline nature of the generated resource may be useful in educational settings where reliable Internet connectivity is not always available.

Vani Rachanakar is therefore presented not as a new speech-synthesis technology, but as a practical application of existing AI capabilities to a specific educational need. The tool is intended as a resource that ESL teachers can try, adapt to their own teaching situations, and use to create customised listening activities with minimal technical effort. Further classroom use and teacher feedback will be used to evaluate its usefulness and identify areas in which the utility can be improved.